

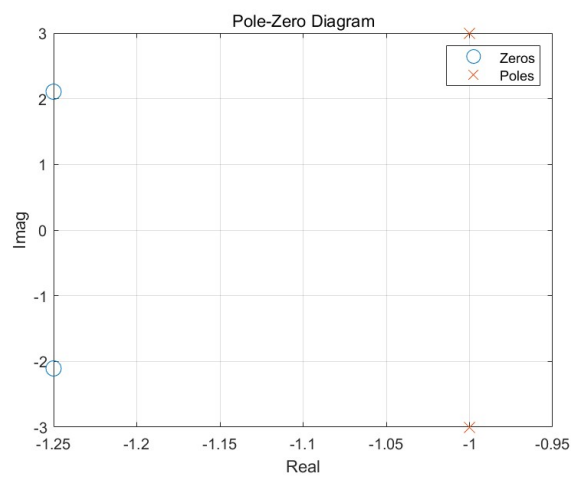
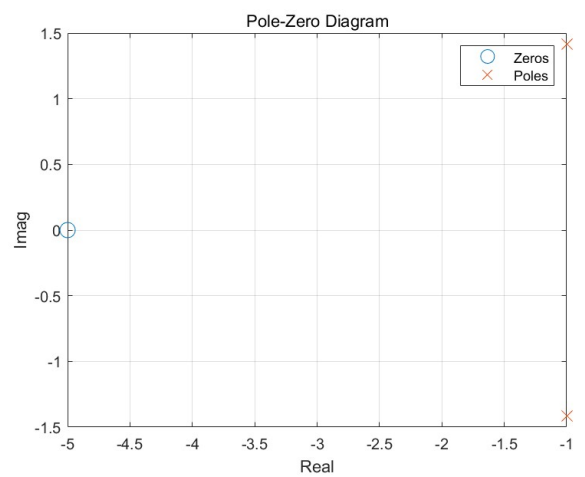
Lab 3

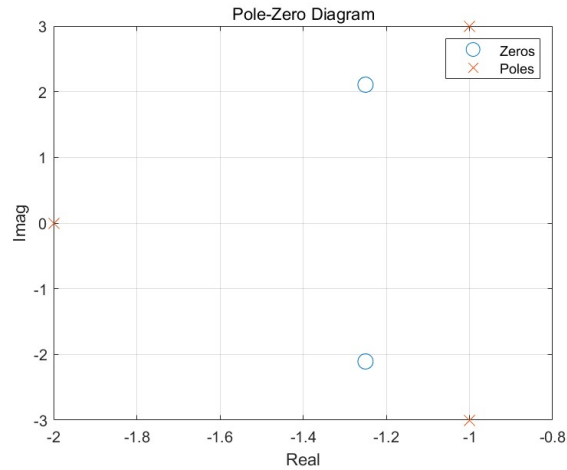
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7 April 2026

Problem 1

(a)





(b)

(i)

Since $H(s) = \frac{s+5}{s^2+2s+3}$, we have $\boxed{\text{zeros}}$ $s = -5$ and $\boxed{\text{poles}}$ $s = -1 \pm \sqrt{2}i$.

for a stable system, ROC must include the entire imaginary axis, so ROC is $\text{Re}(s) > -1$.

(ii)

Since $H(s) = \frac{2s^2+5s+12}{s^2+2s+10}$, we have $\boxed{\text{zeros}}$ $s = \frac{-5 \pm \sqrt{71}i}{4}$ and $\boxed{\text{poles}}$ $s = -1 \pm 3i$.

for a stable system, ROC must include the entire imaginary axis, so ROC is $\text{Re}(s) > -1$.

(iii)

Since $H(s) = \frac{2s^2+5s+12}{(s^2+2s+10)(s+2)}$, we have $\boxed{\text{zeros}}$ $s = \frac{-5 \pm \sqrt{71}i}{4}$ and $\boxed{\text{poles}}$ $s = -1 \pm 3i, s = -2$.

for a stable system, ROC must include the entire imaginary axis, so ROC is $\text{Re}(s) > -1$.

(c)

Diff.eq for the casual LTI system is

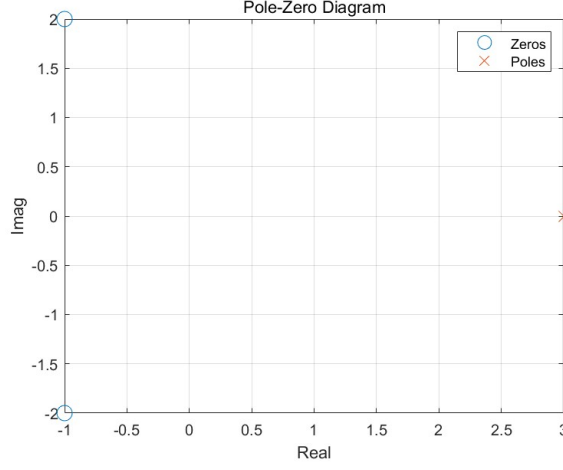
$$\frac{dy(t)}{dt} - 3y(t) = \frac{d^2x(t)}{dt} + 2\frac{dx(t)}{dt} + 5x(t)$$

Laplace Transforms of the diff.eq are

$$sY(s) - 3Y(s) = s^2X(s) + 2sX(s) + 5X(s)$$

or

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s^2 + 2s + 5}{s - 3}$$



we have $\boxed{zeros}$ $s = -1 \pm 2i$ and $\boxed{poles}$ $s = 3$.

(d)

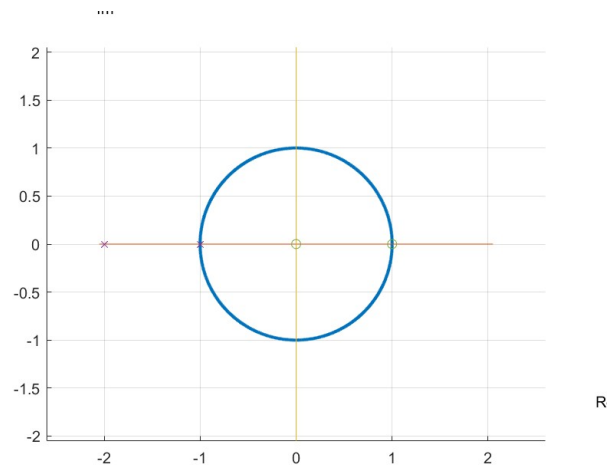
The function `pzplot(b,a,ROC)` (where `ROC` is a point in the ROC) determines the region of convergence as follows:

1. Compute all poles of $H(s)$ using `roots(a)`.
2. Compare the real part of each pole with $\Re(s_0)$, where s_0 is the user-provided `ROC` point.
3. If any poles satisfy $\Re(p_i) < \Re(s_0)$ (poles to the left), find the left boundary as $\ell = \max\{\Re(p_i) \mid \Re(p_i) < \Re(s_0)\}$. Draw a vertical dashed line at $\Re(s) = \ell$.
4. If any poles satisfy $\Re(p_i) > \Re(s_0)$ (poles to the right), find the right boundary as $r = \min\{\Re(p_i) \mid \Re(p_i) > \Re(s_0)\}$. Draw a vertical dashed line at $\Re(s) = r$.
5. The ROC is the vertical strip containing s_0 : $\ell < \Re(s) < r$ (if no poles on a side, that boundary is $\pm\infty$).

This works because the ROC of a rational Laplace transform is always a vertical strip bounded by the real parts of poles and contains no poles.

Problem 2

(a)



(b)

$$y[n] + y[n-1] + 0.5y[n-2] = x[n]$$

Taking the Z-transform of both sides (assuming zero initial conditions):

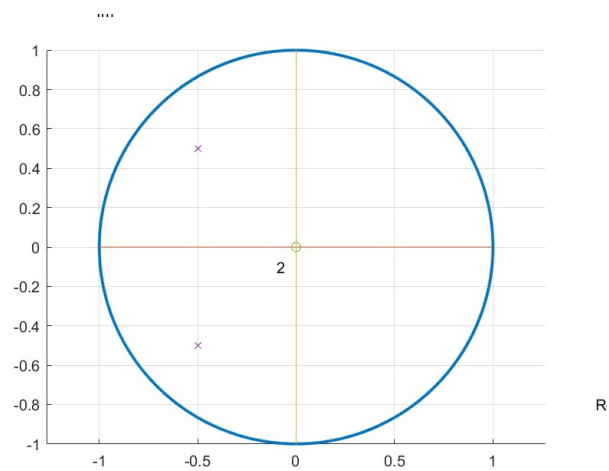
$$Y(z) + z^{-1}Y(z) + 0.5z^{-2}Y(z) = X(z)$$

Factor out $Y(z)$:

$$Y(z) (1 + z^{-1} + 0.5z^{-2}) = X(z)$$

Thus, the system function is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 + z^{-1} + 0.5z^{-2}}$$



(c)

$$y[n] - 1.25y[n-1] + 0.75y[n-2] - 0.125y[n-3] = x[n] + 0.5x[n-1]$$

Taking the Z-transform:

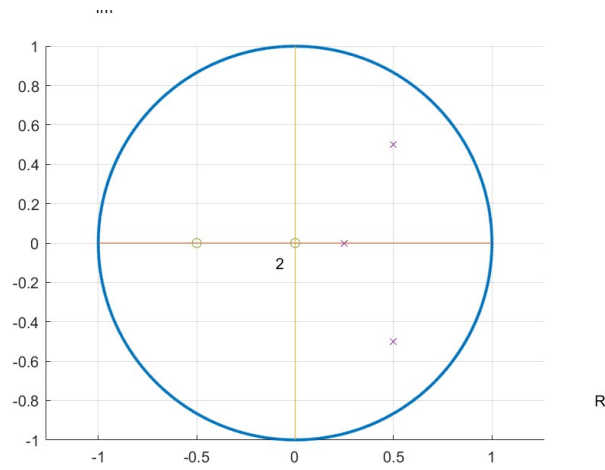
$$Y(z) - 1.25z^{-1}Y(z) + 0.75z^{-2}Y(z) - 0.125z^{-3}Y(z) = X(z) + 0.5z^{-1}X(z)$$

Factor out $Y(z)$ and $X(z)$:

$$Y(z) (1 - 1.25z^{-1} + 0.75z^{-2} - 0.125z^{-3}) = X(z) (1 + 0.5z^{-1})$$

Thus, the system function is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 0.5z^{-1}}{1 - 1.25z^{-1} + 0.75z^{-2} - 0.125z^{-3}}$$



Problem 3

(a)

The original differential equation (9.11) has system function:

$$H_{ac}(s) = \frac{\sum_{m=0}^M b_m s^m}{\sum_{k=0}^K a_k s^k}.$$

The time-reversed differential equation (9.12) (substituting $t \rightarrow -t$) has system function:

$$H_c(s) = \frac{\sum_{m=0}^M (-1)^m b_m s^m}{\sum_{k=0}^K (-1)^k a_k s^k} = \frac{\sum_{m=0}^M b_m (-s)^m}{\sum_{k=0}^K a_k (-s)^k} = H_{ac}(-s).$$

Thus:

- **Pole relationship:** If p is a pole of $H_{ac}(s)$, then $-p$ is a pole of $H_c(s)$. The poles are symmetric about the origin (both real and imaginary parts change sign).
- **System function relationship:** $H_c(s) = H_{ac}(-s)$.

(b)(c)

Diff.eq is

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

The laplace transform is

$$sY(s) + 2Y(s) = X(s)$$

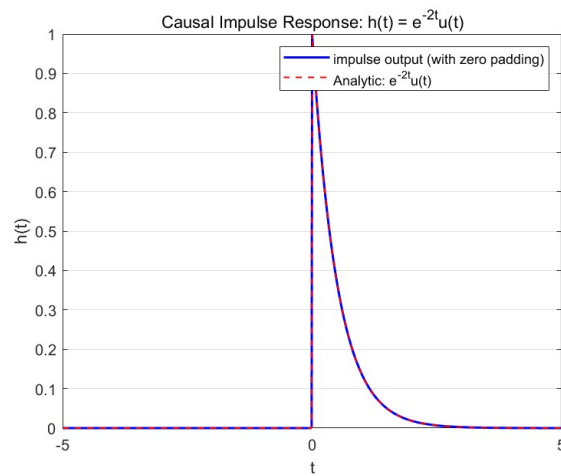
or

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+2}$$

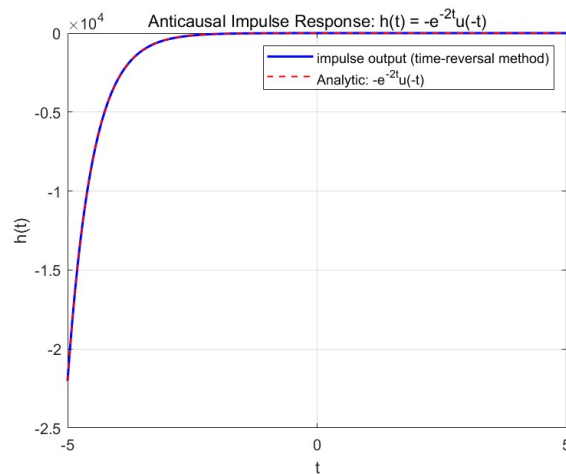
The impulse response is $h(t) = e^{-2t}u(t)$, ROC is $Re(s) > -2$ (casual) .

Or $h(t) = -e^{-2t}u(-t)$, ROC is $Re(s) < -2$ (anticausal) .

(d)



(e)



(f)

Given:

- $h(t) = -e^{-2t}u(-t)$
- $x(t) = e^{5t/2}u(-t)$

The output is $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$.

For $t > 0$, the convolution yields $y(t) = 0$ because the integrand is zero.

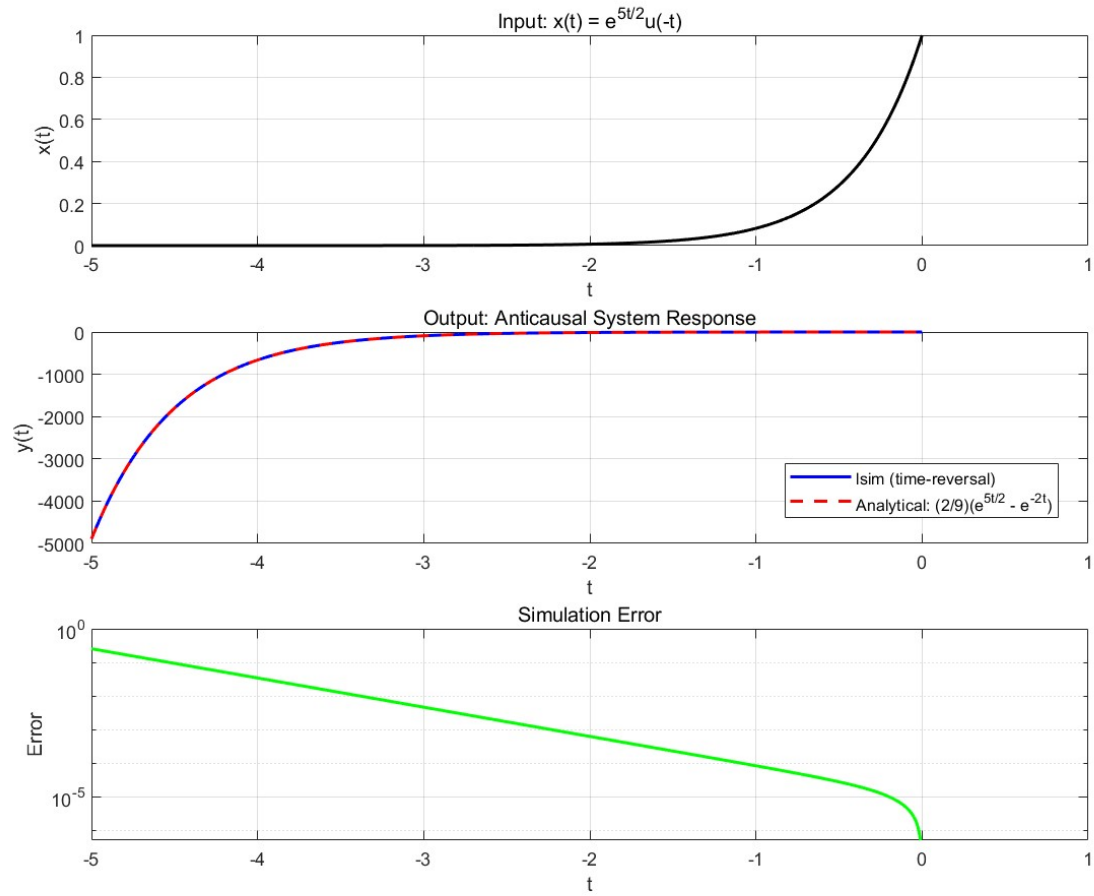
For $t \leq 0$:

$$\begin{aligned}y(t) &= \int_{\tau=t}^0 [-e^{-2\tau}] \cdot [e^{5(t-\tau)/2}] d\tau \\&= -e^{5t/2} \int_t^0 e^{-2\tau-5\tau/2} d\tau \\&= -e^{5t/2} \int_t^0 e^{-9\tau/2} d\tau \\&= -e^{5t/2} \left[-\frac{2}{9} e^{-9\tau/2} \right]_t^0 \\&= -e^{5t/2} \cdot \left(-\frac{2}{9} \right) (1 - e^{-9t/2}) \\&= \frac{2}{9} e^{5t/2} (1 - e^{-9t/2}) \\&= \frac{2}{9} (e^{5t/2} - e^{-2t})\end{aligned}$$

Therefore:

$$\boxed{y(t) = \frac{2}{9} (e^{5t/2} - e^{-2t}) u(-t)}$$

(g)



(h)

The system function is:

$$H(s) = \frac{s^2 + 7s + 21}{s^3 + s^2 + 24s - 26}$$

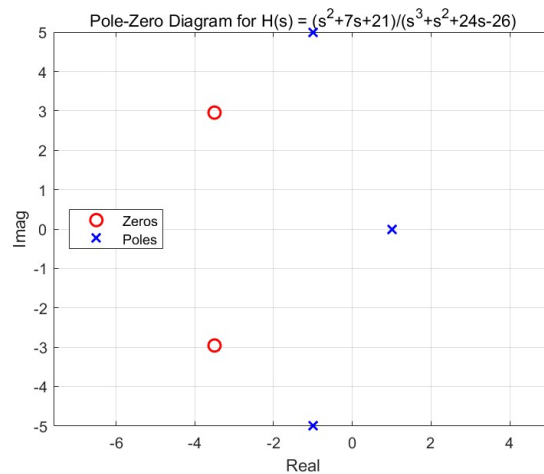
Poles and Zeros:

- Zeros: $s = -3.5 \pm j2.78$ (from $s^2 + 7s + 21 = 0$)
- Poles: $s \approx 1$, $s \approx -1 \pm j5$ (from $s^3 + s^2 + 24s - 26 = 0$)

Possible ROCs:

1. $\Re(s) > 1$ (causal, UNSTABLE)
2. $-1 < \Re(s) < 1$ (noncausal, STABLE)
3. $\Re(s) < -1$ (anticausal, UNSTABLE)

Stable system ROC: $-1 < \Re(s) < 1$ (vertical strip containing the imaginary axis).



(i)

Using **residue**:

$$H(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \frac{r_3}{s - p_3}$$

where $p_1 \approx 1$, $p_{2,3} \approx -1 \pm j5$, and r_i are the corresponding residues.

Impulse responses for each ROC:

- ROC: $\Re(s) > 1$ (causal, unstable):

$$h(t) = (r_1 e^{p_1 t} + r_2 e^{p_2 t} + r_3 e^{p_3 t}) u(t)$$

- ROC: $-1 < \Re(s) < 1$ (stable, noncausal):

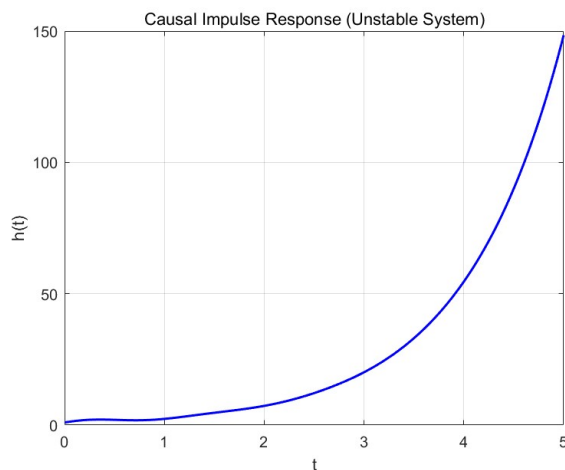
$$h(t) = r_1 e^{p_1 t} u(-t) + (r_2 e^{p_2 t} + r_3 e^{p_3 t}) u(t)$$

- ROC: $\Re(s) < -1$ (anticausal, unstable):

$$h(t) = -(r_1 e^{p_1 t} + r_2 e^{p_2 t} + r_3 e^{p_3 t}) u(-t)$$

(j)

The causal system corresponds to ROC $\Re(s) > 1$. Using **impulse(b,a,ts)**:



Auxiliary conditions for causal system: Initial rest conditions: $y(t) = 0$ and all derivatives zero for $t < t_0$.

(k)

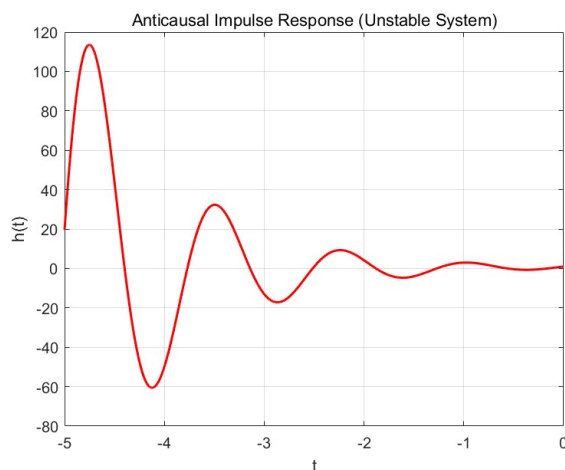
The anticausal system corresponds to ROC $\Re(s) < -1$. Using time-reversal method:

1. Time-reverse the differential equation to obtain a causal system:

$$H_c(s) = H(-s) = \frac{s^2 - 7s + 21}{-s^3 + s^2 - 24s - 26}$$

2. Multiply numerator and denominator by -1 to make leading coefficient positive.
3. Compute impulse response $w(t)$ of the causal system using `impulse`.
4. Time-reverse to get $h_{\text{anticausal}}(t) = w(-t)$.

Auxiliary conditions for anticausal system: Final rest conditions: $y(t) = 0$ and all derivatives zero for $t > t_0$.



(l)

From the partial fraction expansion in part (i):

$$H(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \frac{r_3}{s - p_3}$$

where:

- $p_1 \approx 1$ (right-half plane pole)
- $p_2, p_3 \approx -1 \pm j5$ (left-half plane poles)

For the stable system with ROC $-1 < \Re(s) < 1$, we decompose $H(s) = H_1(s) + H_2(s)$ where:

$$H_1(s) = \frac{r_2}{s - p_2} + \frac{r_3}{s - p_3}, \quad \text{ROC: } \Re(s) > -1$$
$$H_2(s) = \frac{r_1}{s - p_1}, \quad \text{ROC: } \Re(s) < 1$$

Thus $h_1(t)$ is causal and $h_2(t)$ is anticausal.

(m)

$$h_1(t) = r_2 e^{p_2 t} u(t) + r_3 e^{p_3 t} u(t)$$
$$h_2(t) = -r_1 e^{p_1 t} u(-t)$$

The stable system's impulse response is $h_s(t) = h_1(t) + h_2(t)$.

(n)

Combining the causal poles, $H_1(s)$ is a proper rational function with real coefficients:

$$H_1(s) = \frac{b_2 s^2 + b_1 s + b_0}{s^2 + a_1 s + a_0}$$

The corresponding differential equation is:

$$\frac{d^2 y_1(t)}{dt^2} + a_1 \frac{dy_1(t)}{dt} + a_0 y_1(t) = b_2 \frac{d^2 x(t)}{dt^2} + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

Auxiliary conditions: Initial rest conditions. For a second-order system:

$$y_1(t_0^-) = 0, \quad y_1'(t_0^-) = 0$$

(o)

For $H_2(s) = \frac{r_1}{s - p_1}$ with ROC $\Re(s) < p_1$:

$$\frac{dy_2(t)}{dt} - p_1 y_2(t) = r_1 x(t)$$

Auxiliary conditions: Final rest conditions (since the system is anticausal):

$$y_2(t_0^+) = 0$$

(p)

The input signal is:

$$x(t) = \begin{cases} 1, & -3 \leq t \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

For the causal part $H_1(s)$, the output $y_1(t)$ is computed directly using numerical integration with initial rest conditions.

For the anticausal part $H_2(s)$, the output $y_2(t)$ is obtained via time reversal:

1. Define the time-reversed input $r(t) = x(-t)$.
2. The time-reversed system is causal: $H_{\text{rev}}(s) = -\frac{r_1}{s + p_1}$.
3. Compute $w(t)$ as the response of $H_{\text{rev}}(s)$ to input $r(t)$ using initial rest conditions.
4. Obtain $y_2(t) = w(-t)$.

The total output of the stable noncausal system is:

$$y(t) = y_1(t) + y_2(t)$$

The simulation is performed on the interval $-10 \leq t \leq 10$ with time samples. The following plots are generated:

- Input $x(t)$: a rectangular pulse from $t = -3$ to $t = 2$.
- $y_1(t)$: output of the causal subsystem.
- $y_2(t)$: output of the anticausal subsystem.
- $y(t) = y_1(t) + y_2(t)$: total output of the stable noncausal system.

This parallel realization allows numerical simulation of the stable LTI system despite its noncausal nature.

